

Problem 1 [20 points] Linear Regression I

We would like to fit a linear regression estimate to the dataset with X and Y two variables, and $Y = wX$ where w is a constant showing the weight. Given n independent sample pairs, $(x^{(1)}, y^{(1)})$, $(x^{(2)}, y^{(2)})$, ..., $(x^{(n)}, y^{(n)})$, we aim to estimate w by solving the following problem.

$$w^* = \arg \min_w \frac{1}{2} \left(\sum_{i=1}^n (y^{(i)} - wx^{(i)})^2 + \lambda w^2 \right)$$

where $\lambda \geq 0$ is a tuning hyperparameter.

- a) Give a solution in explicit formula for w^* so that w^* can be computed using only the training data and λ . Hint: take the derivative of the objective function and set it to zero.

Take the derivative of the objective function and set it to zero, we have

$$\sum_{i=1}^n x_i(y_i - \beta x_i) + \lambda \beta = 0 \quad (1)$$

$$\hat{\beta} = \frac{\sum_{i=1}^n x_i y_i}{\lambda + \sum_{i=1}^n x_i^2} \quad (2)$$

The closed-form solution of $\hat{\beta}$ (Equation 1) is a monotonic function. When λ increases, the denominator gets larger, and the absolute value of $\hat{\beta}$ decreases, i.e., $\hat{\beta}$ shrinks to zero.

- b) When λ goes from 0 to infinity, how does w^* change? Give a brief explanation of your answer.

The close form of w is a monotonic function, When λ increases, the denominator gets larger, and the absolute value of w decreases (i.e., w shrinks to zero)

Problem 2 [20 points] Linear Regression II

Suppose you know the number of keyboard and mic sold at various locations around the world and from that you want to estimate the number of computers sold using linear regression. Your model is $Y = w_1 k + w_2 m$ where Y is the number of computers sold, k is the number of keyboards sold and m is the number of mice sold. You get 101 observations such that 100 of them have 1 keyboard, 1 mouse, and 1 computer, but the 101st has 1 keyboard, 0 mouse, and 1 computer.

For (a) and (b), you can use `LinearRegression()` in Scikit-learn to compute the answers (turn off the intercept in linear model: `fit_intercept = False`): [link](#)

- a) What are the optimal values of w_1 and w_2 in the scenario above?

$$w_1 = 1 \text{ and } w_2 = 0$$

- b) Now suppose you get two additional observations, both with 0 keyboard, 1 mouse, and 1 computer. What are the optimal w_1 and w_2 values now?

$$w_1 = \frac{1}{3} \text{ and } w_2 = \frac{2}{3}$$

- c) As you should notice, the optimal values for w fluctuate wildly with the addition of even very few observations. Why is this problem happening? Given an arbitrary dataset X, Y , how can we test whether such behavior might occur?

A small number of additional observations cause a huge fluctuation in the parameter coefficients. If our parameters supposedly represent some real-world hidden parameter, then this is not good behavior. This is happening because the parameters are highly correlated. We can compute the correlation for X to check if something like this may be a problem.

Problem 3 [30 points] Logistic Regression

For logistic regression model, we first have a decision function $z = b + w_1x_1 + w_2x_2 + \dots + w_nx_n$; and we have an object assigned to positive example with a probability of $p(y = 1|x) = \sigma(z) = \frac{1}{1+e^{-z}}$.

Consider the following 15 points in a two-dimensional feature space (x_1, x_2) with labels as indicated (green + is positive, red $\times$ is negative):

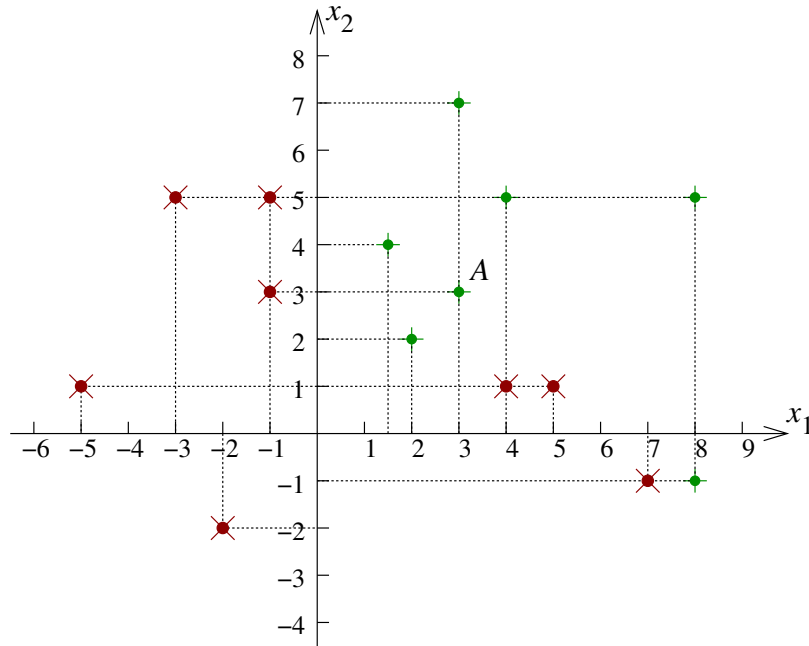


Figure 1: Data set for question 2

Given the decision function

$$z(x_1, x_2) = 4x_1 + 6x_2 - 24$$

answer the following questions:

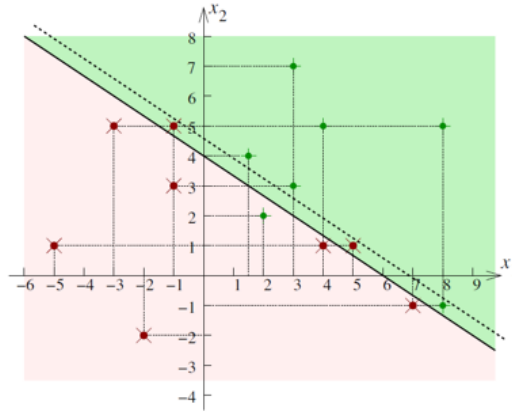
- a) What is the z value of the decision function for point A ? And what's the probability of point A as a positive example?

$$z(3, 3) = 4 \cdot 3 + 6 \cdot 3 - 24 = 12 + 18 - 24 = 30 - 24 = 6$$

The probability of point A as a positive example is

$$p(y = 1|x) = \sigma(z) = \sigma(6) = 0.9975$$

- b) In the figure above, draw the decision boundary, $z(x_1, x_2) = 0$, and indicate the positive and negative regions (half-planes).



The decision boundary is shown with a **solid line**; it passes through points $(0, 4)$ and $(6, 0)$ (you can easily find these by solving the equations $4 \cdot 0 + 6x_2 = 24$ and $4x_1 + 6 \cdot 0 = 24$ respectively). The positive region, $f(x_1, x_2) > 0$, and negative region, $f(x_1, x_2) < 0$, are shown in green and red, respectively.

- c) What is the accuracy of this linear classifier?

Three points are misclassified (1 positive and 2 negative), hence the accuracy is

$$12/15 = 0.8$$

- d) Write the equation for a linear decision function that improves upon this accuracy.

A possible answer is

$$f'(x_1, x_2) := 4.5x_1 + 6x_2 - 27 = 0,$$

which is the same line shifted up by 0.5 units. This yields an accuracy of

$$13/15 = 0.866.$$

Problem 4 [30 points] Model Evaluation

a) What are the possible reasons for model overfitting?

(1) Limited training data size, (2) High model complexity.

b) What can you do to avoid overfitting? Elaborate your answers with **logistic regression**.

(1) Train the model with more data; (2) Use some methods to control model complexity, eg, adding regularizations on logistic regression models, and control the number of nodes in decision tree models.

c) What's the difference between **validation** dataset and **test** dataset?

A valid dataset is a sample of data held back from training your model that is used to give an estimate of model performance while tuning model's hyperparameters. A test dataset that is also held back from the training of the model, but is instead used to give an unbiased estimate of the performance of the final tuned model.

d) Why **accuracy** can be a bad metric to evaluate your classifier? What other metrics you can use?

Accuracy can be a bad metric when the labels (classes) are imbalanced. We can evaluate the classifier by looking at the confusion matrix, which leads to other metrics such as precision, recall, F1 measure.

e) Consider a synthetic data set which is generated from the function $\sin(2\pi x)$, which is shown by the green curve in Figure 2(a). Let us assume that we decided to fit a polynomial function of the following form:

$$y(x, \mathbf{w}) = w_0 + w_1x + w_2x^2 + \dots + w_Mx^M$$

1. Figure 2(b) shows what happens when we fit a linear model ($M = 1$ in the Equation) to the observations. Is this model a good fit? If not, is it underfitting or overfitting?

2. Figure 2(c) shows what happens when we fit a higher order polynomial (degree = $M = 9$ in the Equation) to the observations. Is this model a good fit? If not, is it underfitting or overfitting?

1. Underfitting; 2. overfitting.

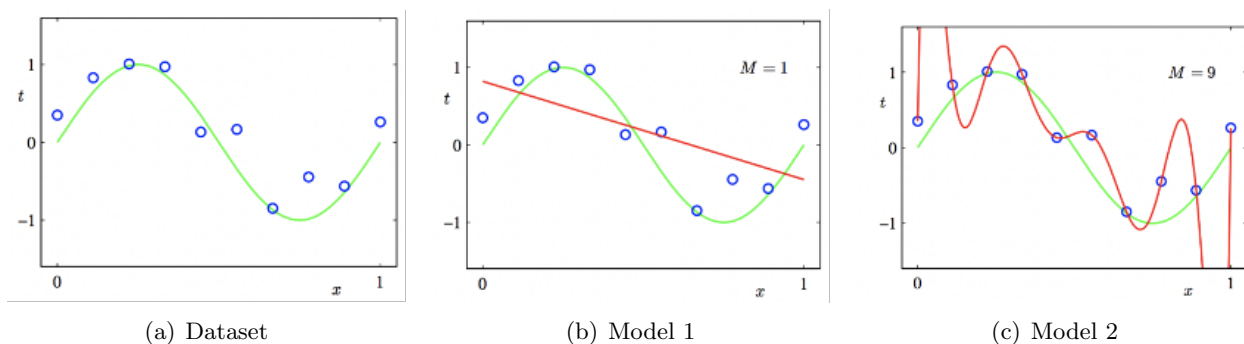


Figure 2: Question 4(e)